Mixed Pattern Test Paper

Section A: Dotted numbering

1. What is the boiling point of water at sea level?
2. What is the chemical symbol for gold?
3. Name the largest planet in the solar system.
4 Define photosynthesis in one sentence.
5. State Newton's second law of motion.

Section B: Long answer questions

Q1. Explain the process of cellular respiration in detail.
Q2. Discuss the causes of the French Revolution.

Problems to solve

Problem 1: Solve for x given 2x + 5 = 15.
Problem 2: Find the derivative of f(x) = x^3 + 2x.
Problem 3: Evaluate the integral of sin(x) dx.

Section C: Multiple choice (no numbering)

What is the capital of France?
A) Paris
B) London
C) Berlin
D) Madrid
Answer: A

Which gas do plants primarily absorb during photosynthesis?
A) Oxygen
B) Nitrogen
C) Carbon dioxide
D) Hydrogen
Answer: C

Section D: Bracketed numbering

1) Describe the structure of an atom.
2) What is the function of mitochondria?

Section E: Roman numeral sub-parts

i. Derive the quadratic formula.
ii. Prove the Pythagorean theorem.
iii. Explain the law of conservation of energy.
iv. State the ideal gas equation.
v. Derive the formula for kinetic energy.